

1. Project Presentation and Context

1.1 Introduction

Tourism is one of the most important economic sectors in many countries. However, planning a trip often requires searching through many different websites to find accommodation, transportation, restaurants, and tourist attractions. This process can be time-consuming and inefficient for tourists.

With the rapid development of web technologies, online platforms have become essential tools to simplify travel planning and improve the tourist experience.

This project aims to design and implement a **collaborative web platform for tourist services** that centralizes essential tourism information and services in a single system.

The platform will allow tourists to explore destinations, search for hotels and restaurants, and make reservations, while service providers such as hotels, restaurants, and travel agencies can manage and promote their services online.

1.2 Problem Statement

Currently, tourists often face difficulties when organizing their trips because tourism services are distributed across multiple platforms and websites. This fragmentation makes it difficult to compare options and plan trips efficiently.

The problem addressed in this project is:

How to create a centralized web platform that allows tourists to easily find and book tourism services while enabling service providers to manage and promote their offers effectively?

1.3 Project Objectives

The main objectives of this project are:

- To design and develop a **web-based platform for tourism services**.
 - To simplify the **trip planning process** for tourists.
 - To provide a **digital space for tourism service providers** to promote their services.
 - To improve **user experience through search, filtering, and booking functionalities**.
 - To encourage **interaction between tourists and service providers** through reviews and ratings.
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2. Functional Requirements

Functional requirements describe the services and features that the system must provide.

2.1 Tourist Module

Tourists represent the main users of the platform. They can interact with the system in several ways.

Tourists should be able to:

- Create an account and log in to the platform.
 - Browse and search tourist destinations and cities.
 - View information about cities including weather and tourist attractions.
 - Search for hotels and restaurants, tourist experience.
 - Apply filters to search results based on:
 - category
 - price
 - rating
 - location
 - available services
 - View detailed information about hotels and restaurants.
 - Make reservations for accommodation or restaurants.
 - Leave reviews and ratings for services they used.
 - View other users' reviews and ratings.
 - Can contact or message the service provider
 - Save favorite places or services for future reference.
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2.2 Service Provider Module

Service providers are professionals such as hotels, restaurants, travel agencies, or tour guides who offer tourism-related services.

Service providers should be able to:

- Register on the platform and create a professional profile.
 - Add new services such as hotel rooms, restaurant menus, or tourist activities.
 - Modify or delete existing offers.
 - Manage the availability of their services.
 - View and manage booking requests from tourists.
 - Communicate with tourists through the platform messaging system.
 - Promote special offers or discounts.
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2.3 Administrator Module

The administrator is responsible for managing and maintaining the platform.

The administrator should be able to:

- Manage user accounts (tourists and service providers).
 - Validate or approve service provider registrations.
 - Monitor platform activity.
 - Manage reviews and remove inappropriate content if necessary.
 - Ensure the proper functioning and security of the system.
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3. Non-Functional Requirements

Non-functional requirements define the quality attributes of the system.

3.1 Usability (User Interface and User Experience)

The platform should provide a simple and intuitive interface that allows users to easily navigate through the system.

The interface must be **responsive**, meaning it adapts to different devices such as:

- desktop computers
 - tablets
 - smartphones.
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3.2 Security

Security is an essential aspect of the system.

The platform must ensure:

- protection of personal user data
 - secure login and authentication
 - restricted access to administrative functions.
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3.3 Performance

The system must provide:

- fast loading times for pages and images
 - efficient search and filtering operations
 - smooth navigation between pages.
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3.4 Availability

The web platform should be available **24 hours a day and 7 days a week**, allowing users to access tourism services at any time.

4. Technical Specifications

4.1 System Architecture

The platform will follow a **Client–Server architecture**.

The system will consist of three main components:

Frontend (Client Side)

The frontend is responsible for displaying the user interface and interacting with the user.

Backend (Server Side)

The backend processes requests, manages business logic, and communicates with the database.

Database

The database stores all the system data such as users, bookings, and services.

4.2 Technologies Used

The following technologies will be used to implement the system.

Frontend technologies:

- React.js

Backend technologies:

- Node.js with Express

Database:

- MySQL
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4.3 Development Tools

Several development tools will be used during the project:

- Visual Studio Code for programming
 - Git for version control
 - Draw.io or Lucidchart for UML diagrams
 - Figma for user interface design
 - MySQL Workbench for database management
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5. Work Methodology

The project development will follow several phases:

1. Requirement analysis and definition of system specifications.
 2. System modeling using **UML diagrams**, including:
 - Use Case Diagram
 - Class Diagram
 - Sequence Diagram
 3. Database design.
 4. Development of the frontend interface.
 5. Development of the backend functionalities.
 6. Integration between frontend, backend, and database.
 7. Testing and validation of the platform.
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7. Expected Results

At the end of the project, the system should provide a functional web platform that:

- allows tourists to easily search and book tourism services
- allows service providers to manage and promote their offers
- provides a user-friendly and responsive interface
- demonstrates the practical application of web development technologies.